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4.1 Modes of continuity

4.2 Modes of convergence

We now begin to talk about different ways a function a sequence of functions f_n can converge to f. This can happen *pointwise* or *uniform*.

Definisjon 1: 4.2.1

Let (X, d_X) and (Y, d_Y) be two metric spaces and let (f_n) be a sequence of functions $f_n : X \to Y$. We say that (f_n) converges pointwise to a function $f : X \to Y$ if $(f_n(x)) \to f(x)\ \forall x \in X$. This means that for each x and each $\varepsilon > 0$ there exists an $N \in \mathbb{N}$ such that $d_Y\left(f_n(x), f(x)\right) < \varepsilon\ \forall n \geqslant N$.

Note that the N in the last sentence depends on x. We may need a much larger N for other x, and much smaller N for other x. If we can use the *same* N for all $x \in X$, we have uniform convergence.

Definisjon 2: 4.2.2

Let (X, d_X) and (Y, d_Y) be two metric spaces and let (f_n) be a sequence of functions $f_n : X \to Y$. We say that (f_n) converges uniformly to a function $f : X \to Y$ if for each $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that if $n \geqslant N$, then $d_Y\left(f_n(x), f(x)\right) < \varepsilon\ \forall x \in X$.

Note the last element in the definition, where we say that $d_Y\left(f_n(x), f(x)\right) < \varepsilon\ \forall x \in X$

Proposisjon 1: 4.2.3

Let (X, d_X) and (Y, d_Y) be two metric spaces and let (f_n) be a sequence of functions $f_n : X \to Y$. For any function $f : X \to Y$ the following are equivalent

- (f_n) converges uniformly to f
- $\sup\left\{d_Y\left(f_n(x), f(x)\right) : x \in X\right\} \to 0$ as $n \to \infty$

Bevis 1: $i) \Rightarrow ii)$

Assume that (f_n) converges uniformly to f. This means that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\forall n \geqslant N\ d_Y\left(f_n(x), f(x)\right) < \varepsilon\ \forall x \in X$. Now pick $\varepsilon > 0$. Since $d_Y\left(f_n(x), f(x)\right) < \varepsilon\ \forall x \in X$, then ε becomes an upper bound. Hence

$$\sup\left\{d_Y\left(f_n(x), f(x)\right) : x \in X\right\} \leqslant \varepsilon\ \forall n \geqslant N$$

And since we have picked an arbitrary $\varepsilon > 0$, there is no reason that we can pick an even smaller $\varepsilon > 0$. Hence

$$\sup\left\{d_Y\left(f_n(x), f(x)\right) : x \in X\right\} \to 0 \text{ as } n \to \infty$$

Bevis 2: $ii) \Rightarrow i)$

We can instead prove $\neg i) \Rightarrow \neg ii)$

Assume that (f_n) does not converge uniformly to f. Then

$$\forall N \in \mathbb{N}\ \exists \varepsilon > 0\ \&\ x' \in X \ \text{ s.t. } \ \forall n \geqslant N \ \text{ then } d_Y\left(f_n(x'), f(x')\right) \geqslant \varepsilon$$

Hence $\sup\left\{d_Y\left(f_n(x'), f(x')\right) : x' \in X\right\} \geqslant \varepsilon$ can impossibly converge to 0 as $n \to \infty$

Bevis 3: $ii) \Rightarrow i)$

Assume that $\sup\{d_Y(f_n(x), f(x)) : x \in X\} \to 0$ as $n \to \infty$. This means that $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\forall n \geqslant N$ and $\forall x \in X$, then $\sup\{d_Y(f_n(x), f(x)) : x \in X\} \leqslant \varepsilon$. But then $d_Y(f_n(x), f(x)) < \varepsilon\ \forall n \geqslant N$ and $\forall x \in X$. Hence (f_n) converges uniformly to f

Proposisjon 2: 4.2.4

Let (X, d_X) and (Y, d_Y) be two metric spaces and assume that (f_n) is a sequence of continuous functions $f_n : X \to Y$ converging uniformally to a function f. Then f is continuous

Bevis 4: 4.2.4

Let $a \in X$. Given $\varepsilon > 0$, we must find a $\delta > 0$ such that when $d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon$.

Since (f_n) converges uniformally, then $\exists N \in \mathbb{N}$ such that $\forall n \geqslant N$ then $d_Y(f_n(x), f(x)) < \varepsilon\ \forall x \in X$. Since f is continuous, its continuous at all $x \in X$, and since $a \in X$, f is continuous at a. This means that $\exists \delta > 0$ such that $d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon$. This means that

$$d_Y(f(x), f(a)) \leqslant d_Y(f(x), f_n(x)) + d_Y(f_n(x), f_n(a)) + d_Y(f_n(a), f(a)) < 3\varepsilon$$

Hence f is continous at $a \in X \forall n \geqslant N$

Oppgave 1: 4.2.1

Let $f_n : \mathbb{R} \to \mathbb{R}$ be defined by $f_n(x) = \frac{x}{n}$. Show that (f_n) converges pointwise, but not uniformly to 0

Bevis 5: 4.2.1

Pick $x' \in \mathbb{R}$. Then $f(x') = \frac{x'}{n}$. We have that

$$\lim_{n\to\infty} f_n(x') = \lim_{n\to\infty} \frac{x'}{n} = 0$$

Hence $f_n(x) \to f(x)$ as $n \to \infty$

To show that f_n is not uniformly convergent, let $\varepsilon := \frac{1}{2}$. Then $\forall n \in \mathbb{N}$, we have that $f_n(n) = 1$. So $\forall n > 0$ we have that

$$|1 - 0| = 1 > \frac{1}{2} = \varepsilon$$

Since $\mathbb{N} \subseteq \mathbb{R}$, then there is a point when $n \in \mathbb{R}$ where (f_n) does not converge to 0 $\forall x \in \mathbb{R}$.

Oppgave 2: 4.2.2

Let $f_n : (0, 1) \to \mathbb{R}$ be defined by $f_n(x) = x^n$. Show that (f_n) converges pointwise but not uniformly to 0.

Bevis 6: 4.2.2

(f_n) converges uniformly to 0 if $\forall \varepsilon > 0 \exists N \in \mathbb{N}$ such that $\forall n \geqslant N\ d(f_n(x), 0) < \varepsilon\ \forall x \in X$. Let $\varepsilon > 0$, then since $x \in (0, 1)$, we get that $\lim_{n\to\infty} x^n = 0$. Hence $\exists N \in \mathbb{N}$ such that $\forall n \geqslant N\ \ |f_n(x) - 0| < \varepsilon$ and (f_n) converges pointwise to 0

To prove that (f_n) is not uniformly convergent to 0, we see that from 4.2.3, uniform convergence is equivalent with

$$\sup\{|f_n(x) - 0| : x \in (0, 1)\} \to 0 \text{ as } n \to \infty$$

We have that $\sup\{x^n : x \in (0, 1)\} = 1^n$. And since, $\lim_{n\to\infty} 1^n = 1 \neq 0$, we have non-uniform converge for (f_n) towards 0.

Oppgave 3: 4.2.5

Let $f_n : \mathbb{R} \to \mathbb{R}$ and assume that the sequence (f_n) of continous functions converges uniformly to $f : \mathbb{R} \to \mathbb{R}$ on all intervals $[-k, k], k \in \mathbb{N}$. Show that f is continuous.

Bevis 7: 4.2.5

Let $k \in \mathbb{N}$. Then $f_n : \mathbb{R} \to \mathbb{R}$ converges uniformly to $f : \mathbb{R} \to \mathbb{R}$ on $[-k, k]$. This means that

$$\sup_{x \in [-k,k]} |f_n(x) - f(x)| \to 0 \text{ as } n \to \infty$$

Since $k \in \mathbb{N}$, we can pick $k := n$ and automatically get that f is continuous on all of $\mathbb{R}$

Bevis 8: Last proof is wrong, corrected

Let $k \in \mathbb{N}$ be such that $x_0 \in [-k, k]$. Since (f_n) is uniformly convergent to f then given $\varepsilon > 0$ $\exists N \in \mathbb{N}$ s.t. $\forall n \geqslant N$ $|f_n(x) - f(x)| < \varepsilon$ $\forall x \in [-k, k]$.

Since (f_n) is a sequence of continuous functions, then $\forall n \in \mathbb{N}$, we have that $\exists \delta > 0$ s.t. $\forall x \in [-k, k]$ $|x - x_0| < \delta \Rightarrow |f_n(x) - f_n(x_0)| < \varepsilon$.

Furthermore, since $(f_n) \to f$ uniformly on $[-k, k]$, then

$$|f_n(x_0) - f(x_0)| \leqslant \sup_{x \in [-k,k]} |f_n(x) - f(x)| < \varepsilon$$

Now show that f is continous at x_0. We are already given $\varepsilon > 0$ and $\delta > 0$, so $\forall n \geqslant N$ we have that when $|x - x_0| < \delta$

$$|f(x) - f(x_0)| \leqslant |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| < 3\varepsilon$$

Now my only question is, isnt $\mathbb{R}$ supposed to be cont on all of $\mathbb{R}$? How to show that if f is continous at a closed set $[-k, k]$, then f is continous at all of $\mathbb{R}$?

Oppgave 4: 4.2.9

Let (X, d) be a metric space and assume that the sequence of continuous functions (f_n) converges uniformly to f. Show that if (x_n) is a sequence converging to x, then $f_n(x_n) \to f(x)$ as $n \to \infty$. Find an example which shows that this is not necessarily the case if (f_n) converges pointwise to f.

Bevis 9: 4.2.9

Let (x_n) be a sequence converging to $x \in X$. Show that $\forall \varepsilon > 0$ $\exists N \in \mathbb{N}$ s.t. $d(f_n(x_n), f(x)) < \varepsilon$ $\forall n \geqslant N$. Since (f_n) converges uniformly to f. This means that forall $n \in \mathbb{N}$ f_n is continuous at x_n. Hence given $\varepsilon > 0$ $\exists \delta > 0$ such that $\forall x \in X, d(x, x_n) < \delta \Rightarrow d(f_n(x_n), f_n(x)) < \varepsilon$

Since f_n converges uniformly to f, then given $\varepsilon > 0$ $\exists N \in \mathbb{N}$ such that $\forall n \geqslant N$ $d(f_n(x), f(x)) < \varepsilon$ $\forall x \in X$

Hence we get that $\forall n \geqslant N$ and given $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$d(f_n(x_n), f(x)) \leqslant d(f_n(x_n), f_n(x)) + d(f_n(x), f(x)) < 2\varepsilon$$

Hence $\exists N \in \mathbb{N}$ such that $\forall n \geqslant N$ we have that $d(f_n(x_n), f(x)) < 2\varepsilon$. And hence $f_n(x_n) \to f(x)$ as $n \to \infty$

Can we find an example that show that this is not necessarily the case if (f_n) only converges pointwise to f?

Oppgave 5: Define not bounded sets

Bevis 10

Let $A \subseteq X$. Then A is bounded if

$$\exists M \in \mathbb{R}\ \forall x, y \in A\ d(x, y) \leqslant M$$

A is not bounded if

$$\forall M \in \mathbb{R}\ \exists x, y \in A \text{ such that } d(x, y) > M$$

Bevis 11

Another definition is $\exists x_0 \in A$ such that $\forall M > 0\ \exists x \in A$with $d(x, x_0) > M$. This holds because of the triangle inuequality

$$M < d(x, y) \leqslant d(x, x_0) + d(x_0, y)$$

Bevis 12

Define $\sup_{x,y \in A} \{d(x, y)\}$, then $A \subseteq X$ of metric space (X, d) is unbounded $\iff \sup_{x,y \in A} \{d(x, y)\} = \infty$

Oppgave 6: Define a subset in metric space is not open

Bevis 13

Let $A \subseteq X$. A is open if

$$\forall x \in A\ \exists \varepsilon > 0 \text{ s.t. } B(x, \varepsilon) \subseteq A$$

Equivilantly, $A \subseteq X$ is not open if

$$\forall \varepsilon > 0\ \exists x \in A \text{ s.t. } B(x, \varepsilon) \nsubseteq A$$

Bevis 14

$A \subseteq X$ is not open if A^c is not closed.

Oppgave 7: Define a subset in metric space is not closed

Bevis 15

Let $A \subseteq X$ be closed, then

$$\forall (x_n) \in A \text{ s.t. } x_n \to x \in X \Rightarrow x \in A$$

Hence A is not closed if

$$\exists (x_n) \in A \text{ s.t. } x_n \to x \in X \nRightarrow x \in A$$

Bevis 16

If A is not closed, then A^c is not open

Oppgave 8: A sequence in a metric space does not converge

Bevis 17

Let $(x_n) \in X$, then (x_n) converges to $x \in X$ if $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon\ \forall n \geqslant N$

(x_n) does not converge if $\forall N \in \mathbb{N} \exists \varepsilon > 0$ such that $\forall n \geqslant N\ d(x_n, x) \geqslant \varepsilon$

Bevis 18

Let (x_n) be a sequence in (X, d), then (x_n) does not converge to $x \in X$, then $x_n \nrightarrow x$ as $n \to \infty$

Oppgave 9: Define a function from one metric space to another is discontinous at a point

Bevis 19

Let (X, d_X) and (Y, d_Y) be two metric spaces. Define $f : X \to Y$. Then f is continuous at $a \in X$ if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \text{ s.t. } \forall x \in X, d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon$$

Then f is discontinous at $a \in X$ if

$$\forall \delta > 0 \; \exists \varepsilon > 0 \text{ s.t. } \exists x \in X \text{ with } d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) \geqslant \varepsilon$$

Oppgave 10: Define a sequence in a metric space is not cauchy

Bevis 20

Let (X, d) be a metric space, and $(x_n) \in X$ a sequence. x_n is a cauchy sequence if $\forall \varepsilon > 0 \; \exists N \in \mathbb{N}$ such that $\forall n, m \geqslant N, d(x_n, x_m) < \varepsilon$

Then (x_n) is not cauchy if $\forall N \in \mathbb{N} \; \exists \varepsilon > 0$ such that $d(x_n, x_m) \geqslant \varepsilon \; \forall n, m \geqslant N$

Oppgave 11: Define a metric space is incomplete

Bevis 21

A metric space is complete if all cauchy sequences converge.

A metric space (X, d) is incomplete if $\exists$ cauchy sequences $(x_n) \in X$ that do not converge

4.3 Integrating and differentiating sequences

Proposisjon 3: 4.3.1

Assume that (f_n) is a sequence of continuous functions converging uniformly to f on $[a, b]$. Then the functions

$$F_n(x) = \int_a^x f_n(t)dt$$

converge uniformly to

$$F(x) = \int_a^x f(t)dt$$

on $[a, b]$.

Bevis 22: 4.3.1

We must show that for a given $\varepsilon > 0$ we can always find an $N \in \mathbb{N}$ such that $|F(x) - F_n(x)| < \varepsilon$ for all $n \geqslant N$ and all $x \in [a, b]$. Since (f_n) converges uniformly to f, there is an $N \in \mathbb{N}$ such that $|f(t) - f_n(t)| < \frac{\varepsilon}{b-a}$ for all $t \in [a, b]$. For $n \geqslant N$, we then have for all $x \in [a, b]$

$$|F(x) - F_n(x)| = \left| \int_a^x (f(t) - f_n(t)) \right| dt \leqslant \int_a^x |f(t) - f_n(t)| \, dt < \int_a^x \frac{\varepsilon}{b-a} dt \leqslant \int_a^b \frac{\varepsilon}{b-a} dt = \varepsilon.$$

This shows that (F_n) converges uniformly to F on $[a, b]$

Korollar 1: 4.3.2

Assume that (f_n) is a sequence of continuous functions converging uniformly to f on the interval $[a, b]$. For any $x_0 \in [a, b]$, the functions

$$F_n(x) = \int_{x_0}^x f_n(t)dt$$

converge uniformly to $F(x) = \int_{x_0}^{x} f(t)dt$ on $[a, b]$.

Bevis 23: 4.3.2

$\int_{x_0}^{x} = \int_{x_0}^{x} f_n(t)dt = \int_a^x f_n(t)dt - \int_a^{x_0} f_n(t)dt$. The first integral converges uniformly to $\int_a^x f_n(t)dt$, and the second integral converges uniformly to $\int_a^{x_0} f_n(t)dt$. Hence $\int_{x_0}^{x} f_n(t)dt$ converges uniformly to

$$\int_a^x f(t)dt - \int_a^{x_0} f(t)dt = \int_{x_0}^{x} f(t)dt.$$

Korollar 2: 4.3.3

Assume (v_n) is a sequence of continuous functions such that the series $\sum_{n=0}^{\infty} v_n(x)$ converges uniformly on the interval $[a, b]$. Then for any $x_0 \in [a, b]$, the series $\sum_{n=0}^{\infty} \int_{x_0}^{x} v_n(t)dt$ converges uniformly and

$$\int_{x_0}^{x} \sum_{n=0}^{\infty} v_n(t)dt = \sum_{n=0}^{\infty} \int_{x_0}^{x} v_n(t)dt$$

Bevis 24: 4.3.3

Assume the series $\sum_{n=0}^{\infty} v_n(x)$ converges uniformly to the function f. This means the partial sums $s_N(x) = \sum_{n=0}^{N} v_k(x)$ converge uniformly to f, and hence by 4.3.2,

$$\int_{x_0}^{x} \sum_{n=0}^{\infty} v_n(t)dt = \int_{x_0}^{x} f(t)dt = \lim_{N\to\infty} \int_{x_0}^{x} s_N(t)dt = \lim_{N\to\infty} \int_{x_0}^{x} \sum_{n=0}^{N} v_n(t)dt = \lim_{N\to\infty} \sum_{n=0}^{N} \int_{x_0}^{x} v_n(t)dt = \sum_{n=0}^{\infty} \int_{x_0}^{x} v_n(t)dt.$$

Proposisjon 4: 4.3.4 Weierstrass M test

Let (v_n) be a sequence of functions $v_n : A \to \mathbb{R}$ defined on a set A, and assume that there are convergent series $\sum_{n=0}^{\infty} M_n$ of nonnegative numbers, such that $|v_n(x)| \leqslant M_n$ for all $n \in \mathbb{N}$ and all $x \in A$. Then the series $\sum_{n=0}^{\infty} v_n(x)$ converges uniformly on A.

Bevis 25: 4.3.4

Let $s_n(x) = \sum_{k=0}^{n} v_k(x)$ be the partial sums of the original series. Since the series $\sum_{n=0}^{\infty} M_n$ converges, we know its partial sums $S_n = \sum_{k=0}^{n} M_k$ form a cauchy sequence. Since for all $x \in A$ and all $m > n$,

$$|s_m(x) - s_n(x)| = \left| \sum_{k=n+1}^{m} v_k(x) \right| \leqslant \sum_{k=n+1}^{m} |v_k(x)| \leqslant \sum_{k=n+1}^{m} M_k = |S_m - S_n|,$$

we see that $(s_n(x))$ is a cauchy sequence in $\mathbb{R}$ for each $x \in A$ and hence converges to a limit $s(x)$. This defines pointwise limit function $s : A \to \mathbb{R}$.

To prove that (s_n) converges uniformly to s, note that for every $\varepsilon > 0$ $\exists N \in \mathbb{N}$ such that if $S = \sum_{k=0}^{\infty} M_k$, then

$$\sum_{k=n+1}^{\infty} M_k = S - S_n < \varepsilon$$

for all $n \geqslant N$. This means that for all $n \geqslant N$,

$$|s(x) - s_n(x)| = \left| \sum_{k=n+1}^{\infty} v_k(x) \right| \leqslant \sum_{k=n+1}^{\infty} |v_k(x)| \leqslant \sum_{k=n+1}^{\infty} M_k < \varepsilon$$

for all $x \in A$, and hence (s_n) converges uniformly to s on A.

Oppgave 12: 4.3.1

Show that $\sum_{n=0}^{\infty} \frac{\cos(nx)}{n^2+1}$ converges uniformly on $\mathbb{R}$

Bevis 26: 4.3.1

Since $\frac{\cos(nx)}{n^2+1} \leqslant \frac{1}{n^2+1}$, then $\sum_{n=0}^{\infty} \frac{\cos(nx)}{n^2+1} \leqslant \sum_{n=0}^{\infty} \frac{1}{n^2+1}$, which converges. Hence, $\sum_{n=0}^{\infty} \frac{\cos(nx)}{n^2+1}$ converges uniformly on $\mathbb{R}$.

Oppgave 13: 4.3.2

Show that $\sum_{n=1}^{\infty} \frac{x}{n^2}$ converges uniformly on $[-1, 1]$

Bevis 27: 4.3.2

Let $x \in [-1, 1]$, then $|x| \leqslant 1$, and hence, $\frac{|x|}{n^2} \leqslant \frac{1}{n^2}$. And because $\lim_{n\to\infty} \frac{1}{n^2} = 0$, we know that the series

$$\sum_{n=1}^{\infty} \frac{|x|}{n^2} \leqslant \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges uniformly on $[-1, 1]$.

Oppgave 14: 4.3.3

Let $f_n : [0, 1] \to \mathbb{R}$ be defined by $f_n(x) = nx\left(1 - x^2\right)^n$. SHow that $f_n(x) \to 0$ for all $x \in [0, 1]$, but that $\int_0^1 f_n(x)dx \to \frac{1}{2}$

Bevis 28: 4.3.3

Since $x \in [0, 1]$, then $\left(1 - x^2\right) < 1$, then $\left(1 - x^2\right)^n \to 0$ as $n \to \infty$. Hence $nx\left(1 - x^2\right)^n \to 0$.

$\int_0^1 nx\left(1 - x^2\right)^n dx = \int_0^1 \frac{-nx}{2x} u^n du = \frac{-1}{2} \int_0^1 nu^n du = \frac{-1}{2} \frac{n}{n+1} \left[1 - x^2\right]_0^1 = \frac{-1}{2} \frac{n}{n+1} \left(-(1)^n\right) = \frac{1}{2} \frac{n}{n+1} \to \frac{1}{2}$ as $n \to \infty$. I have used usub on $1 - x^2$ to integrate.

Oppgave 15: 4.3.5

- a) Show that by summing a geometric series, that

$$\frac{1}{1 - e^{-x}} = \sum_{n=0}^{\infty} e^{-nx}$$

for $x > 0$.

Bevis 29: 4.3.5 a)

- Explain that we can differentiate the series above term by term to get

$$\frac{e^{-x}}{\left(1 - e^{-x}\right)^2} = \sum_{n=0}^{\infty} ne^{-nx}$$

for all $x > 0$.

- Does the series $\sum_{n=1}^{\infty} ne^{-nx}$ converge uniformly on $(0, \infty)$?

4.5 Spaces of bounded, continuous functions

In this section, we shall consider the space of all bounded functions. This is important to prove existence of solutions to differential equations. If (X, d_X) and (Y, d_Y) are metric spaces, a function $f : X \to Y$ is bounded if the set of values $\{f(x) : x \in X\}$ is bounded. This means that if there is a number $M \in \mathbb{R}$ such that $d_Y(f(x), f(y)) \leqslant M$ for all $x, y \in X$. Its equivalent to say that for any $a \in X$, there is a constant M_a such that $d_Y(f(a), f(x)) \leqslant M_a$ for all $x \in X$

Note that if $f, g : X \to Y$ are two bounded functions, then there is a number K such that $d_Y(f(x), g(x)) \leqslant K$ for all $u, v \in X$. To see this, fix a point $a \in X$ and let M_a and N_a be numbers such that $d_Y(f(a), f(x)) \leqslant M_a$ and

$d_Y(g(a), g(x)) \leqslant N_a$ for all $a \in X$. Since the triangle inequality gives

$$d_Y(f(x), g(x)) \leqslant d_Y(f(x), f(a)) + d_Y(f(a), g(a)) + d(g(a), g(x)) \leqslant M_a + d_y(f(a), g(a)) + N_a$$

we can define $K := M_a + d_y(f(a), g(a)) + N_a$

We now let

$$B(X, Y) = \{f : X \to Y : \text{ is bounded } \}$$

be the collection of all bounded functions. We define the metric to be used on this set

$$\rho(f, g) = \sup\{d_Y(f(x), g(x)) : x \in X\}$$

Note that since both $f, g \in B(X, Y)$, then they are both bounded, which means that $\rho(f, g) < \infty$.

Proposisjon 5: 4.5.1

If (X, d_X) and (Y, d_Y) are two metric spaces, then

$$\rho(f, g) = \sup\{d_Y(f(x), g(x)) : x \in X\}$$

defines a metric on $B(X, Y)$.

Bevis 30: 4.5.1

A metric $d(f, g)$ is a metric on a metric space $B(X, Y)$ if

- $\rho(f, g) \leqslant 0$ and $\rho(f, g) = 0 \iff f = g$ for all $f, g \in B(X, Y)$
- $\rho(f, g) = \rho(g, f)$ for all $f, g \in B(X, Y)$
- for all $f, g, h \in B(X, Y)$ we have that $\rho(f, g) \leqslant \rho(f, h) + \rho(h, g)$.

We only prove tha last point

assume $f, g, h \in B(X, Y)$. Then

$$d_Y(f, g) \leqslant d_Y(f, h) + d_Y(h, g).$$

Taking the supremum over this, we get that

$$\rho(f, g) \leqslant \rho(f, h) + \rho(h, g).$$

How do we define convergence in this metric space? Easy, with uniform convergence.

Proposisjon 6: 4.5.2

A sequence (f_n) converges to f in $(B(X, Y), \rho)$ if and only if f_n converges uniformly to f

Bevis 31: 4.5.2

A sequence of functions (f_n) converges to f if and only if

$$\sup\{d_Y(f_n(x), f(x)) : x \in X\} \to 0.$$

This just means that $\rho(f_n, f) \to 0$ as $n \to \infty$, which is what we wanted to show. This comes from proposition 4.2.3. The same is the other way, if we assume that f_n converges uniformly to f, we just go via proposition 4.2.3

Teorem 1: 4.5.3

Let (X, d_X) and (Y, d_Y) be metric spaces and assume that (Y, d_Y) is complete. Then $(B(X, Y), \rho)$ is also complete.

Bevis 32: 4.5.3

Oppgave 16: 4.5.7

For $f \in B(\mathbb{R}, \mathbb{R})$ and $r \in \mathbb{R}$, we define f_r as $f_r(x) = f(x + r)$

- a) Show that if f is uniform continuous, then $\lim_{r \to 0} \rho(f_r, f) = 0$

Bevis 33: 4.5.7 a)

Let f be uniform continuous. Want to show that $\lim_{r\to 0}\rho\left(f_r, f\right) = 0$. That is, show that

$$\sup\left\{\left|f_r(x) - f(x)\right| : x \in X\right\} \to 0 \text{ as } r \to 0.$$

Given $r \in \mathbb{R}$, we have that $\rho\left(f_r, f\right) = \sup\left\{\left|f(x+r) - f(x)\right| x \in X\right\}$. Since f is uniform continuous, then given $\varepsilon > 0$, we can find $\delta > 0$ such that for all $x + r, x \in \mathbb{R}$ we have that $|(x+r) - x| < \delta \Rightarrow |f(x+r) - f(x)| < \varepsilon$. If $|r| < \delta$, we are guaranteed that $|f(x+r) - f(x)| < \varepsilon$ for all $x + r, x \in \mathbb{R}$, because $|(x+r) - x| = |r| < \delta$. Hence, we know that $\rho\left(f_r, f\right) < \varepsilon$. Then we get that when $r \to 0$, we can pick smaller and smaller δ, which implies we can pick smaller and smaller ε, and hence

$$\lim_{r\to 0}\rho\left(f_r, f\right) = 0.$$

- Show that the function g defined by $g(x) = \cos(\pi x^2)$ is not uniformally continuous on $\mathbb{R}$

Bevis 34: 4.5.7 b)

We have that $g(x) = \cos(\pi x^2)$ is uniformly continuous, if for $g_r = g(x+r)$, we have that $\lim_{r\to 0}\rho\left(g_r, g\right) = 0$. Since

$$g_r = g\left(x+r\right) = \cos(\pi\left(x+r\right)^2) \in [-1, 1]$$

And $\lim_{r\to 0}\rho\left(\cos(\pi\left(x+r\right)^2), \cos(\pi x^2)\right) = \lim_{r\to 0}\rho\left(\cos(\pi x^2), \cos(\pi x^2)\right) = 1 \neq 0$. Hence $g(x) = \cos(\pi x^2)$ is not uniformly continuous

- c) Is it true that

$$\lim_{r\to 0}\rho\left(f_r, f\right) = 0$$

for all $f \in B\left(\mathbb{R}, \mathbb{R}\right)$?

Bevis 35: 4.5.7 c)

Obviously, since $g(x) = \cos(\pi x^2) \in B\left(\mathbb{R}, \mathbb{R}\right)$, then the statement that

$$\lim_{r\to 0}\rho\left(f_r, f\right) = 0 \ \forall f \in B\left(\mathbb{R}, \mathbb{R}\right)$$

is false

4.6 Spaces of bounded, continuous functions

As before, we assume that (X, d_X) and (Y, d_Y) are metric spaces, and define

$$C_b\left(X, Y\right) := \left\{f : X \to Y : \text{ f is bounded and continuous }\right\}$$

This set gives us a lot more to work with. Observe that

$$C_b\left(X, Y\right) \subseteq B\left(X, Y\right)$$

and then the metric $\rho\left(f, g\right) = \sup\left\{d_Y\left(f(x), g(x)\right) : x \in X\right\}$ is inherited to the metric space $C_b\left(X, Y\right)$. We make this very crucial observation

Proposisjon 7: 4.6.1

$C_b\left(X, Y\right)$ is a closed subset of $B\left(X, Y\right)$.

Bevis 36: 4.6.1

Its easiest to show that for every sequence $(f_n) \in C_b(X,Y)$ converging to $f \in B(X,Y)$, then $f \in C_b(X,Y)$. From proposition 4.5.2, we have that for a sequence (f_n) converes to $f \in (B(X,Y), \rho)$ if and only if it converges uniformly to f. From proposition 4.2.4, we have that a sequence of continuous functions (f_n) converging uniformly to a funtion f, then f is also continuous.

For our arbitrary sequence of functions (f_n), we have found it converges uniformly to f, and that f is also continuous. Then $f \in C_b(X,Y)$, and for every convergent sequence $(f_n) \in C_b(X,Y)$ converging to $f \in B(X,Y)$, then f is also continuous, which implies that $C_b(X,Y)$ is closed.

Teorem 2: 4.6.2

Let (X, d_X) and (Y, d_Y) be metric spaces and assume (Y, d_Y) is complete, then $(C_b(X,Y), \rho)$ is also complete

Bevis 37: 4.6.2

Recall 3.4.4 that a closed subset of a complete space is itself complete. Since $B(X,Y)$ is complete from theorem 4.5.3, and $C_b(X,Y)$ is a closed subset of $B(X,Y)$ by the proposition above, it follows that $C_b(X,Y)$ is also complete

The reason we have so far restricted ourselves to the space $C_b(X,Y)$ of bounded continuous functions, is that the supremum

$$\rho(f,g) = \sup\{d_Y(f(x), g(x)) : x \in X\}$$

can take infinite values if f and g are assumed to be only continuous. AS a metric is not allowed to take infinite values, it leads to problems in our theory. The solution is to also restrict ourselves to bounded and continous functions. This problem doesnt occur when X is compact

Proposisjon 8: 4.6.3

Let (X, d_X) and (Y, d_Y) be metric spaces, and assume X is compact, then all continuous functions from X to Y are bounded

Bevis 38: 4.6.3

Assume that $f : X \to Y$, and pick an arbitrary $a \in X$. It suffices to show that $h(x) = d_Y(f(x), f(a))$ is continuous. By the inverse triangle inequality, we get that

$$|d_Y(f(x), a) - d_Y(a, f(y))| \leqslant d_Y(f(x), f(y))$$

and since f is continuous, so is h, and any δ that works for f, will also work for h

If we define

$$C(X,Y) = \{f : X \to Y : \text{ f is continuous }\}$$

the proposition above tells us that for compact X, the spaces $C(X,Y)$ and $C_b(X,Y)$ coincide. In most applications, we shall use $[a,b]$ and we shall then just be working with the space $C(X,Y)$. The following theorem sums up the result above for compact X

Teorem 3: 4.6.4

Let (X, d_X) and (Y, d_Y) be metric spaces, and assume that X is compact, then

$$\rho(f,g) = \sup\{d_Y(f(x), g(x)) : x \in X\}$$

defines a metric on $C(X,Y)$. If (Y, d_Y) is complete, then so is $(C(X,Y), \rho)$

Oppgave 17: 4.6.1

Let $X, Y = \mathbb{R}$. Find $f, g \in C(X,Y)$ such that

$$\rho(f,g) = \infty$$

Bevis 39: 4.6.1

Let $f(x) = \frac{1}{x}$ and let $g(x) = 1$, then

$$\sup\left\{d_Y\left(\frac{1}{x}, 1\right) : x \in \mathbb{R}\right\} = \infty.$$

Hence, I have found $f, g \in C(X, Y)$ such that $\rho(f, g) = \infty$.

Oppgave 18: 4.6.2

Assume $X \subseteq \mathbb{R}^n$ is not compact. Show that there is an unbounded, continuous function $f : X \to \mathbb{R}$.

Bevis 40: 4.6.2

Since X is not compact, then there exists subsequences of $(x_n) \in X$ such that $(x_{n_k}) \to x \in \mathbb{R} \backslash X$. Also, since $X \subseteq \mathbb{R}^n$ is not compact, its also either not closed, not bounded, or both not closed and not bounded. We are supposed to show that a function $f \in C(X, \mathbb{R})$ is not bounded.

We deal with the first case, that f is unbounded. Then f is automatically unbounded and continuous, if we select $f(x) = x$, which blows up as $x \to \infty$. This is allowed, because we deal with the case where f is unbounded.

We deal with the second case, that X is not closed. This means that $\exists (x_n) \to x \notin X$. So there exists a limit point $p \notin X$. Let $(x_n) \in X$ be a sequence, and assume that $x_n \to p$. Then for the function $f(x) = \frac{1}{\|x_n - p\|}$, we have that

$$f(x_n) = \frac{1}{\|x_n - p\|} \to \infty$$

as $x_n \to p$.

Hence, f is also unbounded if X is not closed.

Oppgave 19: 4.6.3

Assume that $f : \mathbb{R} \to \mathbb{R}$ is a bounded continuous function. If $u \in C([0, 1], \mathbb{R})$, we define $L(u) : [0, 1] \to \mathbb{R}$ to be the function

$$L(u)(t) := \int_0^1 \frac{1}{1 + t + s} f(u(s))\, ds$$

- a) Show that L is a function from $C([0, 1], \mathbb{R})$ to $C([0, 1], \mathbb{R})$.

Bevis 41: 4.6.3 a)

We want to prove that L is continuous in t. So we must for $\varepsilon > 0$ find a $\delta > 0$ such that $|t - t_0| < \delta \Rightarrow |L(u)(t) - L(u)(t_0)| < \varepsilon$ for all $t \in [0,1]$. We first assume that $|t - t_0| < \delta$. Then

$$|L(u)(t) - L(u)(t_0)| = \left| \int_0^1 \left(\frac{1}{1+t+s} - \frac{1}{1+t_0+s} \right) f(u(s))\, ds \right|.$$

This is

$$\int_0^1 \left| \frac{t - t_0}{(1+t+s)(1+t_0+s)} \right| f(u(s))ds \leqslant \int_0^1 |t - t_0| f(u(s))ds$$

Since f is a bounded continuous function, we know that $\exists M \in \mathbb{R}$ such that $f(u(s)) \leqslant M$ for $s \in [0,1]$. So

$$\int_0^1 |t - t_0| f(u(s))ds \leqslant M \cdot |t - t_0| \int_0^1 ds = M \cdot |t - t_0| < M\delta.$$

So if we pick $\delta := \frac{\varepsilon}{M}$, then we get that when $|t - t_0| < \delta$, then $|L(u)(t) - L(u)(t_0)| < \varepsilon$, and thus L is continuous. Hence, L is a function that takes in a continuous function, and spits out another continuous function.

- Assume that

$$|f(u) - f(v)| \leqslant \frac{C}{ln(2)} |u - v|$$

for all $u, v \in \mathbb{R}$ for some number $C < 1$. Show that the equation $Lu = u$ has a unique solution in $C([0,1], \mathbb{R})$.

Bevis 42: 4.6.3 b)

This looks a lot like a contraction. If we prove that L is a contraction, then we can use Banachs fixed point theorem to state that $Lu = u$ has a unique solution in $C([0,1], \mathbb{R})$. But we first have to prove that $C([0,1], \mathbb{R})$ is complete. This happens if all cauchy sequences converge. I will make this proof below. For now, assume that $C([0,1], \mathbb{R})$ is complete.

For a $t \in [0,1]$, we have that

$$|Lu(t) - Lv(t)| \leqslant \int_0^1 \frac{1}{1+t+s} |f(u(s)) - f(v(s))|\, ds \leqslant \frac{C}{ln2} \int_0^1 \frac{1}{1+t+s} |u(s) - v(s)|\, ds$$

Since $u : [0,1] \to \mathbb{R}$, we have that $|u(s) - v(s)| < \rho(u,s)$ for all $s \in [0,1]$. Hence

$$\begin{aligned} \frac{C}{ln2} \rho(u,v) \int_0^1 \frac{1}{1+t+s} &= \frac{C}{ln2} \rho(u,v) \cdot ln\left(\frac{2+t}{1+t} \right) \\ &\leqslant \frac{C}{ln2} \rho(u,v)\, ln2 = C\rho(u,v) \end{aligned}$$

Observe now that

$$\sup\{|Lu(t) - Lv(t)| : t \in [0,1]\} \leqslant \sup\{|C\rho(u,v)| : t \in [0,1]\} = C\rho(u,v)$$

then we get that $\rho(Lu, Lv) \leqslant C\rho(u,v)$. And since $C < 1$, we get that L is a contraction. And by Banachs fixed point theorem, there exists a unique point in $C([0,1], \mathbb{R})$ such that $Lu = u$

Bevis 43: Proof that $C([0,1], \mathbb{R})$ is complete

A metric space is complete if all cauchy sequences converge. Let (f_n) be a cauchy sequence in $C([0,1], \mathbb{R})$. By definition, $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ such that $\forall n, m \geqslant N\ \rho(f_n, f_m) < \varepsilon$.

We have for all $x \in [0,1]$ that $|f_n(x) - f_m(x)| \leqslant \rho(f_n, f_m)$, so $f_n(x)$ is a cauchy sequence of real numbers. Since

$\mathbb{R}$ is complete, we can define

$$f(x) := \lim_{n\to\infty} f_n(x).$$

Thus $f : [0,1] \to \mathbb{R}$ is a well defined point wise limit of f_n.

Now to deal with the uniform convergence. Fix $n \geqslant N$. For any $x \in [0,1]$ and $m \geqslant N$, then

$$|f_n(x) - f_m(x)| < \varepsilon.$$

Let $m \to \infty$ and get that $|f_n(x) - f(x)| < \varepsilon \ \forall x \in [0,1]$. This is the definition of uniform convergence for a sequence. So $\rho(f_n, f) \to 0$ as $n \to \infty$. Hence all cauchy sequences converge, and $C([0,1], \mathbb{R})$ is complete.